Zero Emissions Vehicles in California

Your Name Here

Replace the placeholders below with your responses and R code. Refer to the project assignment for the full instructions. For Part 4, complete four of the five questions and remove the section you do not choose.

Part 1: The CEC Dashboard

Write three different questions you are curious about that can be answered by interacting with the CEC dashboard.

1. [Your first question here.]
2. [Your second question here.]
3. [Your third question here.]

Part 2: The Data Behind the Dashboard

Import the data

Save `New_ZEV_Sales_Last_updated_07-17-2026_ada.xlsx` in your repository's `data/` folder. Add code below to load the packages you need and import the County sheet as `zev`.

```
# Add your package-loading and data-import code here.
```

What does the data record?

After reviewing all three sheets and the dashboard documentation, explain precisely whether the data records new ZEV sales, purchases, or registrations.

[Your answer here.]

Included and excluded transactions

What types of transactions are included in the tally, and what types are excluded?

[Your answer here.]

Unit of observation

For the County sheet, what does each row represent?

[Your answer here.]

Part 3: Checking the Dashboard

For each statistic, identify where it appears on the dashboard and recreate it from `zev` using a `dplyr` pipeline. Report your result and compare it with the dashboard. If you cannot recreate a statistic, explain why; remove any unused code chunk.

1. Cumulative light-duty ZEV sales through 2026
Dashboard location and reported value(s): [Your response here.]

```
# Add your dplyr pipeline here.
```

Result and comparison, or explanation of why this cannot be recreated:

[Your response here.]

2. Year-to-date 2026 total sales
Dashboard location and reported value(s): [Your response here.]

```
# Add your dplyr pipeline here.
```

Result and comparison, or explanation of why this cannot be recreated:

[Your response here.]

3. Year-to-date 2026 total sales for your county
Use your home county if you are from California; otherwise, use Alameda County.

County: [Your county here.]

Dashboard location and reported value(s): [Your response here.]

```
# Add your dplyr pipeline here.
```

Result and comparison, or explanation of why this cannot be recreated:

[Your response here.]

4. 2026 total sales separated by BEV, PHEV, and FCEV
Dashboard location and reported value(s): [Your response here.]

```
# Add your dplyr pipeline here.
```

Result and comparison, or explanation of why this cannot be recreated:

[Your response here.]

5. Year-to-date ZEV Share
Dashboard location and reported value(s): [Your response here.]

```
# Add your dplyr pipeline here.
```

Result and comparison, or explanation of why this cannot be recreated:

[Your response here.]

6. Counts of 2026 Tesla sales, split by model
Dashboard location and reported value(s): [Your response here.]

```
# Add your dplyr pipeline here.
```

Result and comparison, or explanation of why this cannot be recreated:

[Your response here.]

7. Counts of 2026 sales, split first by make, then by model
Optional extra challenge: replicate the dashboard's row arrangement.

Dashboard location and reported value(s): [Your response here.]

```
# Add your dplyr pipeline here.
```

Result and comparison, or explanation of why this cannot be recreated:

[Your response here.]

Part 4: Beyond the Dashboard

Select four of the following five questions. For each selected question, include your analysis and state the strongest reason someone could argue that your analysis is misleading.

Questions selected: [List four letters here.]

A. Your Questions
Select one of the questions you wrote in Part 1 and answer it using the data. If it cannot be answered, explain what additional data you would need.

Selected question: [Your question here.]

```
# Add your analysis code here, if applicable.
```

Answer, or additional data needed: [Your response here.]

Strongest reason this analysis could be misleading: [Your response here.]

B. Tax Credit Effect

The assignment describes incentives ending on September 30, 2025. Is there a Q3 2025 surge followed by a Q4 drop, and does it differ between BEVs and PHEVs?

```
# Add your analysis code here.
```

Answer and comparison of BEVs and PHEVs: [Your response here.]

Strongest reason this analysis could be misleading: [Your response here.]

C. Market Concentration

Compute a Herfindahl index across makes for each year. Is the ZEV market becoming more or less concentrated?

```
# Add your analysis code here.
```

Answer and interpretation: [Your response here.]

Strongest reason this analysis could be misleading: [Your response here.]

D. Brand Geography

Which make is disproportionately popular in each county compared with its statewide share? Explain how you construct your statistic and justify any data exclusions.

Statistic and any exclusions: [Your response here.]

```
# Add your analysis code here.
```

Answer and interpretation: [Your response here.]

Strongest reason this analysis could be misleading: [Your response here.]

E. The Elon Effect

Is there evidence that Tesla sales dropped in left-leaning counties relative to right-leaning counties following Elon Musk's involvement in DOGE in early 2025? Use the county groups provided in the assignment.

```
# County groups provided in the assignment.
dem_counties <- c("Alameda", "San Francisco", "Santa Clara", "Marin",
                  "San Mateo", "Los Angeles", "Santa Cruz", "Humboldt")
rep_counties <- c("Kern", "Shasta", "Modoc", "Lassen", "Tehama", "Kings",
                  "Placer", "El Dorado")
```

Comparison years and rationale: [Your response here.]

Choice of raw sales counts or proportion of ZEV sales, and what this statistic does and does not control for: [Your response here.]

```
# Add your analysis code here.
```

Answer and interpretation: [Your response here.]

Strongest reason this analysis could be misleading: [Your response here.]